

Question:
From the given options, select the most suitable one to fill in the question mark, so that a certain pattern is presented:

Hint:
To solve this pattern transformation puzzle, observe **how the lines in each figure change**. In each step, lines slanting to the left are rotating clockwise and increasing in number, while lines slanting to the right are also rotating clockwise but decreasing in number. Apply this pattern to transition from the third figure to the next, focusing on the number and orientation of the lines to determine the correct configuration.

Question:
From the four given options, choose the most suitable one to fill in the question mark, so that it presents a certain regularity:

Hint:
Think about the pattern formed by the intersecting shapes in the first set of figures. **Notice how the number of sides of the intersection area increases sequentially** with each figure. Apply this pattern to the second set of figures to determine the shape of the intersection at the question mark. Which option continues this sequence?

Question:
Choose the most appropriate option from the given four choices to fill in the question mark so as to present a certain regularity:

Hint:
Examine how the black squares within the grid move. **Focus on the pattern of movement in both the inner and outer parts of the grid**. Note that in the inner 3x3 grid, except for the center square which remains fixed, the other two black squares shift positions in a specific direction and sequence. In the outer grid, observe how the black squares shift consistently in a particular direction. Which option continues this pattern?

Question:
Among the following options, which one conforms to the pattern change of the given figure is:

Hint:
Consider **how the elements of each figure are composed**. Observe that in the sequence on the left, the third figure is a combination of parts from the first two figures. Specifically, the bottom half of the third figure is identical to the bottom half of the first figure, and the top half is identical to the top half of the second figure. Apply this pattern to the sequence on the right. Identify the parts of figures on the right first.

Question:
From the given four options, select the most appropriate one to fill in the question mark so that a certain pattern is formed:

Hint:
Focus on **counting specific shapes within each figure**. Pay attention to how many of a particular geometric shape appears consistently across the options. Remember, some elements like shading may be designed to distract you, so concentrate on the count of the shapes themselves.

Question:
Divide the six figures below into two categories, making sure each category of figures has its own common features or patterns:

Hint:
Consider the problem as one involving grouping and classification. **Focus on the characteristics of the shapes provided**. Notice that all shapes are divided into distinct closed regions. While the number of these regions varies, a more subtle criterion is at play. Pay attention to the largest region in each shape. Examine its symmetry: does it exhibit rotational or reflective symmetry?

Question:
From the four given options, choose the most appropriate one to fill in the question mark so that it presents a certain pattern:

Hint:
Consider the arrangement and symmetry of the elements. **Focus on the properties of the shapes and their composition**. In the first group, the shapes exhibit axial symmetry and are composed of letters. Analyze the letters in both groups, noting the symmetry and repetition. When examining the options, identify the one that maintains the same symmetrical properties and letter composition as the examples given.

Question:
Choose the most appropriate figure from the four given options to fill in the question mark, so that a certain pattern is formed.

Hint:
Consider the symmetry of the curves in the given problem. **Look for an axis of symmetry and check if the curves in the options share this same property**. Additionally, observe whether the curves pass through the origin, as this could be a crucial differentiator. Focus on these attributes to identify the correct option.

Figure 19: Examples of hint prompts. Hint prompts are provided to guide reasoning without revealing the final answer directly.

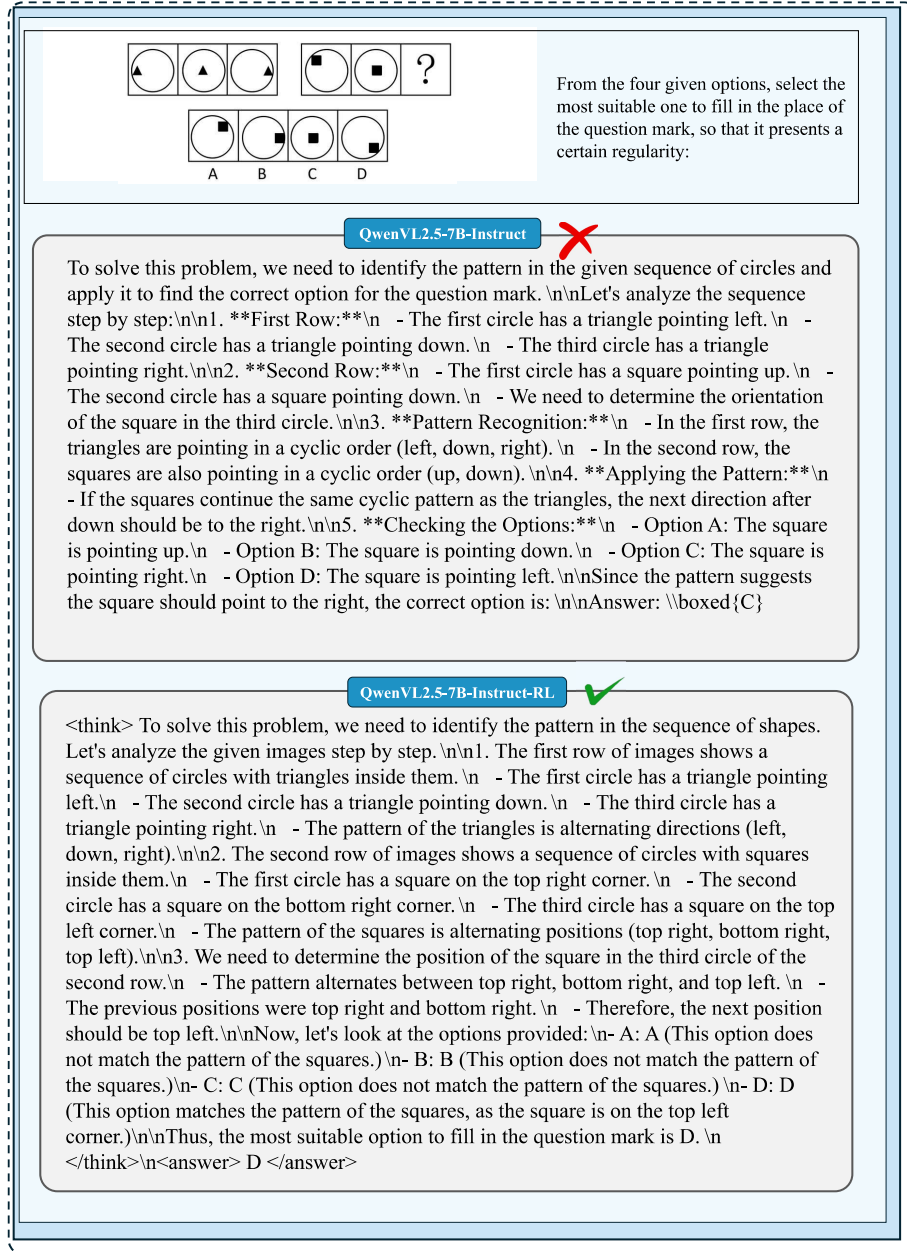


Figure 20: Comparison of model outputs before and after RL training stage for Qwen2.5-VL-7B.

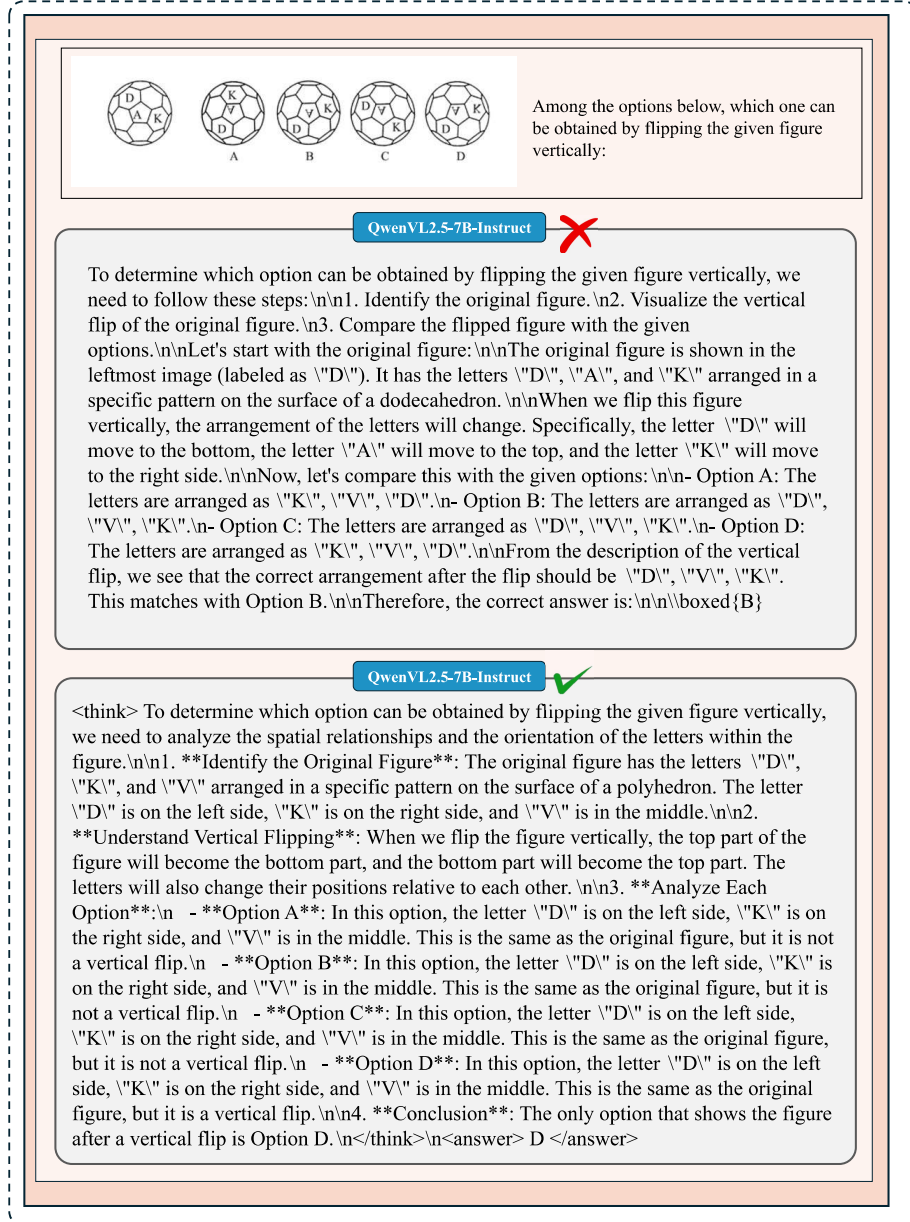


Figure 21: Comparison of model outputs before and after RL training stage for Qwen2.5-VL-7B.